Aalborg University

Department of Architecture, Design and Media Technology

Medialogy

Vitual Reality Prism Effect

Group 1 — 8. Semester project

Authors

Markus Birch Flensborg
Mathias Bisgaard
Rikke Bragh Jensen
Tze Huo Gucci Ho

Supervisors

Bastian Ilsø Hougaard

Title: Virtual Reality Prism Effect
Project: 8. Semester project
Project period: February 2026 - May 2026

AALBORG UNIVERSITY
STUDENT REPORT

Project group:

Markus Birch Flensborg

mflens22@student.aau.dk

Mathias Bisgaard

mbisga22@student.aau.dk

Rikke Bragh Jensen

rbje22@student.aau.dk

Tze Huo Gucci Ho

tho22@student.aau.dk

Abstract:

Supervisors: Bastian Ilsø Hougaard
Department: Department of Architecture, Design and Media Technology

Due date: May 28th, 2026

Number of pages: ..
Appendix: ..

Keywords: Prism Effect; Prism Adaptation; Virtual Reality; C#; Unity;

THE CONTENT OF THIS REPORT IS FREELY AVAILABLE, BUT PUBLICATION (WITH SOURCE INDICATION) MAY ONLY BE MADE IN AGREEMENT WITH THE AUTHORS.

Contents

1 Introduction **1**

2 Background Research **2**
2.1 Spatial neglect in stroke victims . . . 2
2.2 Prism therapy . . . 2
2.3 Prism therapy in virtual reality . . . 3
2.4 State of the Art for Prism Therapy in Virtual Reality . . . 4
2.4.1 Comparing Conventional Prism Therapy to a Virtual Reality Approach with Open-Loop Pointing . . . 4
2.4.2 Virtual prism adaptation therapy via virtual-hand displacement and neuro-physiological data collection . . . 4
2.4.3 Visuomotor Adaptation as Spatial Recalibration with Virtual-Hand Rotation Deviation . . . 5
2.4.4 Virtual Hand Displacement for Reproducing PA without Prisms . . . 5
2.4.5 Multisensory Re-orientation of Attention in Immersive Rehabilitation . . . 5
2.4.6 Literature Review of Virtual Reality Treatment of Unilateral Spatial Neglect Using Eye-Tracking Technology . . . 6
2.5 Problem statement . . . 9

3 Methodology **10**
3.1 Design rationale . . . 10
3.1.1 Independent variables: induction of prism adaptation . . . 10
3.1.2 Procedure: exposure and feedback . . . 10
3.1.3 Behavioural measures . . . 11
3.1.4 Participants . . . 12
3.1.5 Apparatus and implementation . . . 12

4 Design **13**

5 Implementation **14**

6 Evaluation **15**

7 Results **16**

8 Discussion **17**

9 Conclusion **18**

A Design **20**

B Code **21**

C Evaluation **22**

Chapter 1

Introduction

Chapter 2

Background Research

2.1 Spatial neglect in stroke victims

Unilateral spatial neglect (USN) is a common consequence following a stroke and is characterised by a reduced ability to detect, respond to, or orient toward stimuli presented on the contralesional side of the brain lesion[8]. The disorder reflects a deficit in spatial attention and spatial representation rather than a primary sensory or motor impairment. Consequently, patients may neglect and fail to acknowledge objects, people, or even their own limbs on the affected side, reflecting deficits across both egocentric and allocentric spatial representations, significantly interfering with activities of daily living (ADL) and impeding recovery. Because USN affects both perceptual and motor aspects of spatial behaviour, rehabilitation methods must target the pathological attentional bias toward the ipsilesional side rather than isolated sensory or motor symptoms. To evaluate these deficits, researchers typically rely on behavioural measures, which are observable performance on spatial tasks that probe different components of spatial processing. Common examples include line bisection, which assesses perceived midline; landmark task, which measures spatial judgement of pre-bisected lines; and open-loop pointing, which evaluates perceived straight-ahead without visual feedback.[1]

2.2 Prism therapy

A well-established method for rehabilitating unilateral spatial neglect is prism adaptation therapy, in which patients typically perform repeated goal-directed visuomotor tasks while wearing laterally displacing wedge prism glasses. The prisms are configured to shift the visual field toward the ipsilesional side of the brain lesion, thereby creating a systematic mismatch between visual input and proprioceptive-motor representations. During task performance, this visuomotor mismatch initially produces expected pointing errors, but continued exposure over repeated trials eventually elicits automatic, error-driven sensorimotor recalibration through repeated movement corrections. Upon removal of the prisms, a compensatory after-effect emerges in the direction opposite to the optical displacement. Because the initial displacement is ipsilesional, the after-effect is expressed toward the contralesional side of space, counteracting the pathological spatial bias characteristic of neglect. Importantly, the therapeutic efficacy of prism adaptation depends not only on this low-level sensorimotor recalibration as reflected in the after-effect, but also on its transfer to higher-order spatial processing and activities of daily life.[3, 10] Standard prism therapy procedures generally involve the following three phases:

1. A pre-exposure phase where baseline visuomotor performance is assessed on a rehabilitation task prior to any prism effect being introduced. This phase establishes the patient's initial spatial and motor accuracy, providing a reference for evaluating subsequent prism adaptation.

2. An exposure phase where patients perform repeated goal-directed actions under prism exposure, during which the prism effect introduces a systematic visuomotor mismatch. Initial pointing errors occur due to this mismatch, where repeated movements eventually lead to

gradual error reduction as the sensorimotor system undergoes recalibration to compensate for the visuomotor mismatch.

3. A post-exposure phase in which prism exposure ends and the prisms are removed. The resulting after-effect of sensorimotor recalibration is then evaluated, and visuomotor performance is reassessed on the same rehabilitation task used in the pre-exposure phase. This phase also examines the transfer of the after-effect and sensorimotor recalibration to higher-order spatial processes.[3, 10]

Prism adaptation (PA) refers to the combined process encompassing error reduction and sensorimotor recalibration during prism exposure, as well as the subsequent after-effect and its transfer once the prisms are removed.[3, 10]

2.3 Prism therapy in virtual reality

Rapid advances in immersive technologies have positioned virtual reality (VR) as a promising clinical tool for neurorehabilitation by enabling an engaging, controlled, and highly configurable therapeutic environments. In recent years, VR has been explored as a method not only for facilitating prism therapy but also for supporting a wider range of interventions for spatial neglect rehabilitation. VR-based implementations of prism therapy generally follow the same procedure as conventional prism therapy, but where the prism effect is simulated digitally rather than produced through physical wedge prism glasses.[9][2][5]

Compared to conventional prism therapy, VR offers several potential strengths. First, immersive head-mounted displays can reduce external visual and auditory distractions while providing a controlled visual environment. Second, VR allows the perturbation to be implemented and adjusted digitally, enabling precise control over the magnitude and direction of the induced prism effect. Third, the virtual environment allows task parameters such as target position, timing, and feedback to be adjusted in detail while simultaneously enabling streamlined data collection. These capabilities allow the full prism adaptation process to be monitored on a trial-by-trial basis rather than relying solely on pre-post measurements. For example, immediate visual feedback of the pointing endpoint provides a direct and quantifiable measure of motor accuracy, allowing the temporal evolution of error reduction and sensorimotor recalibration to be analysed throughout prism exposure. The controlled nature of VR environments may also reduce awareness of the spatial manipulation, which has been suggested to increase the likelihood of successful adaptation in some studies.[9][2][5]

Despite these advantages, VR-based prism therapy also presents potential limitations. Firstly, VR technology can inherently induce cybersickness symptoms in some users, which may cause discomfort and potentially reduce therapeutic efficacy[7]. In addition, the sensorimotor experience in VR differs from real-world interactions due to factors such as tracking accuracy, display latency, and differences in depth perception. As a result, it remains important to establish whether VR-induced perturbations produce prism adaptation comparable to those observed with physical prism glasses.[9][2][5]

Several studies have examined this question by comparing prism adaptation tasks performed with traditional prism glasses and in immersive VR environments. Studies have indicated that VR can induce adaptation in a comparable or even larger proportion of participants while producing after-effects of similar magnitude to conventional prism therapy[9][5]. At the neural level, immer-

sive VR-based prism therapy has also been associated with activation of cortical areas involved in sensorimotor processing and attentional control. Together, these findings suggest that VR implementations can reproduce key aspects of conventional prism adaptation, supporting their use as a controllable platform for inducing and measuring prism-adaptation-related spatial recalibration.[2]

2.4 State of the Art for Prism Therapy in Virtual Reality

The current state of the art in virtual reality implementations of prism therapy present a wide variety of approaches. Across the literature, implementations differ in (1) how the perturbation is generated (e.g., optical prisms, virtual hand offset or rotation, environment/target manipulations), (2) how feedback is presented (e.g., terminal feedback only or continuous feedback; explicit error visualisation), and (3) which which tasks were employed for performance assessment (e.g., open-loop pointing, line or landmark bisection, head-orientation metrics). To provide a detailed overview of current approaches, a structured literature review was conducted. The following sections summarise its key findings, with the full review available in Appendix X.

2.4.1 Comparing Conventional Prism Therapy to a Virtual Reality Approach with Open-Loop Pointing

Anan et al. directly compared conventional prism glasses with a virtual reality-based prism adaptation (VR-PA) system employing an open-loop pointing test in both conditions, making their study a useful foundation for understanding what VR-PA means operationally. They fixed the deviation to 20 diopters (approximately $11.3°$) rightward in both the prism-glasses and VR conditions, and used a repeated reaching protocol of 96 reaches to three targets arranged in an arc[1]. For the conventional prism glasses, reaching was performed from below a table towards tabletop targets, paced by a metronome every 3 seconds[1]. In the VR condition (Oculus Rift + Touch controllers; Unity), the same structure was recreated with three spherical targets in an arc; every 3 seconds one target changed colour as a cue, and participants pressed a controller button where they believed the target was[1]. The VR system explicitly visualised endpoint error (e.g., different colours depending on whether the reach overlapped the target), and the authors emphasise the advantage in VR which enables fine-grained adjustment of the deviation angle in 1 diopter increments and logging of performance data[1]. Their operational definition of successful adaptation was a leftward shift on the open-loop pointing test after exposure.[1]

2.4.2 Virtual prism adaptation therapy via virtual-hand displacement and neurophysiological data collection

Cho et al. developed an immersive virtual prism adaptation therapy (VPAT) system that induces the prism effect by shifting the virtual hand relative to the real hand during repeated pointing, explicitly aiming to replicate conventional prism therapy without physical prisms[2]. In their feasibility study with 14 healthy adults, the protocol consisted of four sequential phases: pre-VPAT, VPAT-$10°$, VPAT-$20°$, and post-VPAT[2]. The key prism adaptation signature is described in similar terms as conventional prism therapy: early exposure shows rightward pointing errors under the imposed deviation, errors decrease with repeated pointing (error reduction), and a leftward after-effect appears in the post-exposure phase when the deviation is removed[2]. Methodologically, the study's distinct contribution is that it couples the behavioural protocol to functional near-infrared spectroscopy (fNIRS), where they analyse changes in oxyhaemoglobin concentration during pointing across phases and report that the most significantly activated channels during pointing contrasts

were located in the right hemisphere, with candidate regions including dorsolateral prefrontal cortex and frontal eye field[2]. Their interpretation is that VPAT can induce behavioural adaptation and may modulate attention-related networks while providing an instrumented way to quantify errors in VR.[2]

2.4.3 Visuomotor Adaptation as Spatial Recalibration with Virtual-Hand Rotation Deviation

Gerken et al. explicitly frame prism adaptation in VR within the broader concept of visuomotor adaptation: a motor-learning process driven by sensory prediction errors under a systematic mismatch between intended movement and observed feedback[5]. In their VR task, the perturbation is implemented as a $40°$ rotation of visual feedback during repeated reaching movements[5]. Rather than varying the deviation magnitude across tasks, the manipulation targets the format of feedback, which is an embodied virtual hand versus an abstract cursor, to test whether embodiment and the associated sense of agency changes the adaptation process[5]. Behaviourally, they report that hand-based feedback speeds up early error reduction, whereas after-effects and transfer measures are not altered by feedback type, which they interpret as evidence that the underlying recalibration mechanism is not determined by visual representation alone[5]. This study is representative of a class of VR approaches that induce spatial recalibration via visuomotor rotation (a controlled mismatch in feedback mapping) rather than a shift in the visual field.

2.4.4 Virtual Hand Displacement for Reproducing PA without Prisms

Kim et al. propose that the key component to PA is the repeated coupling of movement to systematically displaced visual feedback, not the optical prism device itself[9]. Their VR approach induces the prism effect by applying an angular deviation between the tracked physical hand and the displayed virtual hand during repeated pointing, such that exposure produces systematic reaching errors that reduce over time and a measurable after-effect once the deviation is removed[9]. In this interpretation, VR is primarily a platform for parametrising the deviation (e.g., selecting the deviation magnitude and logging trial-by-trial error) while keeping the broader visual environment stable[9]. This aligns with later VR-PA comparisons that treat prism effect as an imposed mapping error between action and visual feedback rather than a global shift of the visual scene.[1, 2]

2.4.5 Multisensory Re-orientation of Attention in Immersive Rehabilitation

Heyse et al. demonstrate a qualitatively different direction. Rather than inducing PA through a visuomotor discrepancy and measuring after-effects, they design a multisensory VR tool intended to re-orient attention and exploration in patients with unilateral spatial neglect[6]. Their system uses a xylophone-like instrument where key placement can be changed per task. For assessment, keys are arranged symmetrically and for training, keys are placed off-centre towards the neglected side as a bottom-up manipulation intended to draw attention into contralesional space[6]. Over a two-week clinical period, sessions occurred three times per week (Monday/Wednesday/Friday), lasting approximately 30 minutes and consisting of an Assessment followed by Scales, Memory, and Free-to-Play tasks[6]. Behaviour is quantified through task performance measures (e.g., timing/accuracy on keys) and head-direction metrics, with the authors discussing how stimuli presentation and task difficulty adjustments influence exploration towards the neglected side[6]. This work shows that immersive VR has also been used to target neglect through multisensory cueing and task-driven exploration.[6]

2.4.6 Literature Review of Virtual Reality Treatment of Unilateral Spatial Neglect Using Eye-Tracking Technology

A related work by Kaiser et al. provides a comprehensive literature review of how virtual reality and eye-tracking technologies have been applied in the treatment of unilateral spatial neglect. Their analysis highlights that, although VR and gaze-based interaction offer promising avenues for creating controlled, adaptive rehabilitation environments, current research remains heavily focused on severe or clinically obvious cases of USN. As a result, milder and more heterogeneous forms of neglect, such as subtle attentional biases or deficits that manifest only under specific task demands, are frequently overlooked. Kaiser et al. emphasize that this gap limits the ecological validity and clinical reach of existing VR-based interventions, as these milder USN subtypes are both common and clinically relevant. They therefore call for future work to develop more sensitive assessment methods and design VR tasks capable of addressing the full spectrum of USN presentations. Their findings underline a need for flexible and configurable VR rehabilitation frameworks that can accommodate individual differences in neglect profiles and support targeted therapeutic strategies.[8]

The state-of-the-art review of virtual reality implementations of prism therapy revealed substantial inconsistencies in how researchers describe, simulate, and induce the prism effect and prism adaptation. These inconsistencies extend to terminology, implementation strategies, and task selection. As a consequence, each research group has developed its own interpretation of what the prism effect is, how it should be simulated, and what prism adaptation entails, and has typically implemented a virtual environment specifically tailored to those interpretations. While some methodological elements overlap across studies, these VR environments are generally designed in isolation, vary in structure and parameters, and are seldom built to accommodate different subtypes of unilateral spatial neglect (USN). In practice, this results in research groups reinventing VR solutions for their specific studies, with limited reusability.
This absence of a shared language and framework risks limiting comparability across studies, complicates reproducibility, and restricts the broader applicability of VR-based prism therapy in both research and clinical practice.
Based on these observations, two key research gaps are identified:

1. Lack of a shared terminology: Core concepts such as prism effect and prism adaptation are interpreted and defined differently across studies, complicating interpretability and comparability of findings.

2. Absence of a unified framework: Current VR implementations are largely project-specific and mutually incompatible, offering no standardized or configurable platform for simulating different prism effects, supporting diverse rehabilitation tasks, or accommodating research on USN subtypes. As a result, existing systems vary considerably in structure and functionality, limiting their reusability and comparability across studies.

Collectively, these gaps highlight the need for a shared framework to support future work in VR-based prism therapy. A unified and configurable platform, grounded in consistent terminology and methodology, would streamline experimentation for rehabilitating USN, allow controlled simulation of different prism effects, and support a broader range of rehabilitation tasks. Such a platform

could also enable clinicians and researchers to tailor interventions to specific USN subtypes in a systematic and comparable way.

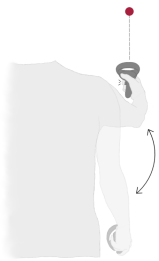
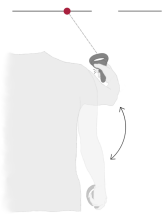
Icon	**Name**	**Description**
	Open Loop Pointing	Open Loop Pointing is...
	Line Bisection	Line Bisection
	Landmark Bisection	Landmark Bisection

Table 2.1: Task Icons

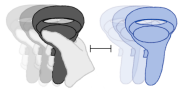
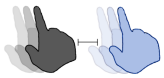
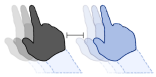
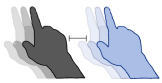
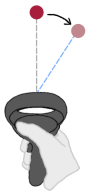
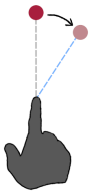
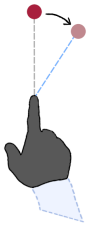
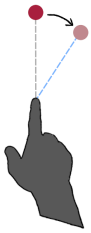
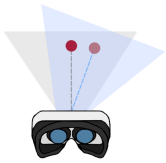
Icon	Name	Description
	Translate - Controller	Translate - Controller
	Translate - Hand	Translate - Hand
	Translate - Hand invisible arm	Translate - Hand invisible arm
	Translate - Arm	Translate - Arm
	Rotation - Controller	Rotation - Controller
	Rotation - Hand	Rotation - Hand
	Rotation - Hand invisible arm	Rotation - Hand invisible arm
	Rotation - Arm	Rotation - Arm
	Skewness	Skewness

Table 2.2: Prism effect icons

2.5 Problem statement

Prism therapy is a well-established method for rehabilitating unilateral spatial neglect (USN). Conventional prism therapy setups rely on physical wedge prism glasses and manually administered tasks, which limits experimental flexibility, standardization, scalability, and addressing of different subtypes of USN. Recent research has shown that virtual reality (VR) can effectively be used to create a controllable environment for prism therapy to simulate prism effects and induce prism adaptation for USN rehabilitation. However, existing VR implementations are typically developed for single specific experiments and lack the modularity necessary for enabling broader research or clinical use. To address this gap, we propose the development of a virtual reality sandbox platform that enables researchers and clinicians to design, configure, and employ a wide range of prism effects and rehabilitation tasks, as well as enabling adaptation of interventions to specific USN subtypes. This platform will aim to support experimentation, facilitate adaptable interventions for USN subtypes, and provide a unified environment for prism therapy in VR. In addition, this study will systematically investigate how different combinations of prism effects and rehabilitation tasks influence the induction of prism adaptation in healthy participants. By evaluating which configurations most effectively elicit prism adaptation, this study aims to gain insights that can guide future therapeutic design. For this purpose, this project will be guided by the following problem statement:

"How do different VR-based methods for inducing spatial recalibration influence the magnitude and characteristics of behavioural adaptation in healthy participants? OR
How do different implementations of the prism effect in VR affect the magnitude of prism adaptation (including error reduction and after-effect)?"

To support this problem statement and further guide the design and implementation of this sandbox system, a set of research questions were formulated:

1. How does the prism adaptation implemented in VR affect healthy people in goal-directed pointing tasks?
2. To what extent does the adaptation transfer to spatial tasks that do not require the trained movement?

Chapter 3
Methodology

The purpose of the experimental design is to compare different VR-based methods for inducing prism adaptation and to evaluate the resulting after-effects using consistent and measurable outcome variables. The methodology therefore separates the induction of visuomotor mismatch from the procedures used to measure adaptation.

3.1 Design rationale

The reviewed literature shows that prism-adaptation-like effects can be induced in immersive environments by introducing a systematic mismatch between executed movement and visual feedback. Studies using virtual hand displacement, visually shifted environments, and visuomotor rotation report comparable behavioural signatures in the form of trial-by-trial error reduction during exposure and a compensatory after-effect after removal of the perturbation [9, 1, 2, 5].
To ensure methodological consistency across conditions, only induction methods that are expected to produce measurable error reduction and after-effects are included. Approaches that primarily train spatial exploration without generating a sensorimotor realignment are therefore not part of the core comparison.

3.1.1 Independent variables: induction of prism adaptation

Spatial recalibration is induced by manipulating the relation between the physical movement of the participant and its visual representation in VR. The sandbox implements the following visuomotor mismatch conditions:

- **Visually shifted environment**: a lateral displacement of the visual scene corresponding to an optical deviation similar to that of traditional wedge prism glasses. This condition reproduces the classical prism effect in an immersive environment while allowing precise control of the deviation magnitude[1].
- **Virtual hand displacement**: a constant lateral offset between the tracked real hand and the rendered virtual hand. This implementation follows VR-based prism-adaptation paradigms in which a positional mismatch produces gradual error reduction and a post-exposure after-effect [9, 1].
- **Visuomotor rotation**: an angular rotation of the visual feedback relative to the executed movement direction. This manipulation is used to provide a reference condition for comparing different reference frames of adaptation[5].

In addition, the magnitude of the imposed deviation can be introduced progressively across trials. Gradual increases in prism strength have been used in dose-response studies to investigate the temporal development of adaptation and are therefore implemented as a controllable parameter[4].

3.1.2 Procedure: exposure and feedback

Participants perform repeated goal-directed pointing movements towards visual targets. During the exposure phase, one of the prism effect conditions is active.

Two feedback modes are considered:

- **Concurrent feedback**, in which the movement is visible throughout its execution.
- **Terminal feedback**, in which only the endpoint of the movement is shown.

Terminal feedback prevents online correction and is commonly used in prism therapy to promote error-driven realignment, whereas concurrent feedback allows continuous visual guidance of the movement.

3.1.3 Behavioural measures

Behavioural measures are assessed using tasks that capture changes in perceived straight-ahead, spatial bias, and trial-by-trial pointing performance. These measures are selected because they are widely used in prism-adaptation studies and allow direct comparison with previous work.

Pointing error during exposure

During the exposure phase, the angular deviation between the target position and the movement endpoint is recorded for each trial. The reduction of this error across trials is used as a measure of the rate of adaptation.

Open-loop pointing

Open-loop pointing is used to measure the after-effect. Participants point towards their perceived straight-ahead without visual feedback of the moving hand. The angular deviation from the true midline indicates the magnitude and direction of the sensorimotor realignment.

Line bisection task

In the line bisection task, participants are presented with a horizontal line and are asked to indicate its perceived midpoint. The deviation from the true centre provides a measure of spatial bias in the internal representation of space. Changes between pre- and post-exposure performance are interpreted as transfer of the adaptation beyond the trained movement.

Landmark task

In the landmark task, participants judge whether a pre-bisected horizontal line is transected to the left or right of its true midpoint. Because the response does not require a pointing movement, this task provides a perceptual measure of spatial bias independent of motor execution.

Error reduction during exposure

For each trial, the angular deviation between the target position and the movement endpoint is recorded. The rate of error reduction across trials represents the speed of adaptation.

After-effect

Following the exposure phase, open-loop pointing is used to assess the shift in perceived straight-ahead. The angular deviation from the true midline is taken as a measure of after-effect magnitude.

Transfer to spatial tasks

To evaluate whether the adaptation influences spatial representation beyond the trained movement, perceptual spatial tasks are administered immediately after exposure. Changes in spatial bias between pre- and post-exposure measurements indicate transfer of the recalibration.

3.1.4 Participants

The experiment is conducted with healthy adult participants. This allows controlled comparison of induction methods without the variability associated with clinical populations and is consistent with previous VR-based prism-adaptation studies[1, 5].

3.1.5 Apparatus and implementation

The study is implemented in an immersive virtual reality system with tracked hand or controller input. The sandbox architecture enables real-time manipulation of the spatial relation between physical movement and visual feedback and ensures that all induction methods are tested within the same virtual environment and task structure.

Chapter 4

Design

Chapter 5

Implementation

Chapter 6

Evaluation

Chapter 7

Results

Chapter 8

Discussion

Chapter 9

Conclusion

Bibliography

[1] Kimika Anan et al. "A Comparative Study of Prism Adaptation Using Virtual Reality and Conventional Prism Glasses." In: *Cureus* 17.9 (2025), e91378–e91378.

[2] Sungmin Cho et al. "Feasibility study of immersive virtual prism adaptation therapy with depth-sensing camera using functional near-infrared spectroscopy in healthy adults". In: *Scientific reports* 12.1 (2022), p. 767.

[3] Francesca Frassinetti et al. "Long-lasting amelioration of visuospatial neglect by prism adaptation". In: *Brain* 125.3 (2002), pp. 608–623.

[4] Roberto Gammeri et al. "Adaptation to virtual prisms and its relevance for neglect rehabilitation: a single-blind dose-response study with healthy participants". In: *Neuropsychological Rehabilitation* 30.4 (2020), pp. 753–766. DOI: 10.1080/09602011.2018.1502672. URL: https://doi.org/10.1080/09602011.2018.1502672.

[5] Johanna Gerken et al. "Sense of ownership is linked to the speed of visuomotor adaptation in virtual reality but not to generalization, intermanual transfer, or aftereffects". In: *Scientific Reports* 15.1 (2025), p. 18690.

[6] Joris Heyse et al. "From patient to musician: a multi-sensory virtual reality rehabilitation tool for spatial neglect". In: *Applied Sciences* 12.3 (2022), p. 1242.

[7] Interaction Design Foundation - IxDF. *Cybersickness in Virtual Reality*. Jan. 2024. URL: https://www.interaction-design.org/literature/topics/cybersickness-in-virtual-reality.

[8] Alexander Pilgaard Kaiser et al. "Virtual Reality and Eye-Tracking Assessment, and Treatment of Unilateral Spatial Neglect: Systematic Review and Future Prospects". In: *Frontiers in Psychology* Volume 13 - 2022 (2022). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2022.787382. URL: https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.787382.

[9] Won-Seok Kim, Nam-Jong Paik, and Sungmin Cho. "Development and validation of virtual prism adaptation therapy". In: *2017 International Conference on Virtual Rehabilitation (ICVR)*. IEEE. 2017, pp. 1–2.

[10] Andrea Serino et al. "Mechanisms underlying neglect recovery after prism adaptation". In: *Neuropsychologia* 44.7 (2006), pp. 1068–1078.

Appendix A

Design

Appendix B

Code

Appendix C

Evaluation